

Towards robust surrogate models: Benchmarking machine learning approaches to expediting phase field simulations of brittle fracture

PFM-Fracture Lite

We introduce the **Lite** version of the PFM-Fracture dataset¹. This version is designed to be more accessible and computationally efficient than the original dataset, following community feedback regarding the complexity of the original dataset focusing on biaxial tension boundary condition and the volumetric-deviatoric energy decomposition.

1 Dataset Specifications

The dataset is partitioned into three sub-datasets based on the topological complexity of the initial conditions.

- **Boundary Conditions:** All samples are simulated under biaxial tension.
- **Energy Decomposition:** Volumetric-deviatoric decomposition.
- **Sample Count:** Each sub-dataset contains 500 samples (1,500 samples total).

1.1 Sub-Dataset Overview

The primary distinction between the sub-datasets is the number of initial cracks, which dictates the complexity of the resulting crack patterns (see Figure 1).

Table 1: PFM-Fracture Lite Dataset: Sub-dataset DOI and Configuration Reference

Sub-Dataset ID	Initial Cracks	Resolution	DOI (Link)
tension-vol-128-lite-1c	1	128	doi.org/10.7910/DVN/J9QVSD
tension-vol-128-lite-2c	2	128	doi.org/10.7910/DVN/DTKMZK
tension-vol-128-lite-3c	3	128	doi.org/10.7910/DVN/NRZZRC
tension-vol-fullres-lite-1c	1	Full Res	doi.org/10.7910/DVN/HKXBMO
tension-vol-fullres-lite-2c	2	Full Res	doi.org/10.7910/DVN/GIHUIZ
tension-vol-fullres-lite-3c	3	Full Res	doi.org/10.7910/DVN/NMAMIO

¹Hamdi, Erfan, and Emma Lejeune. "Towards robust surrogate models: Benchmarking machine learning approaches to expediting phase field simulations of brittle fracture." Computer Methods in Applied Mechanics and Engineering 449 (2026): 118526.

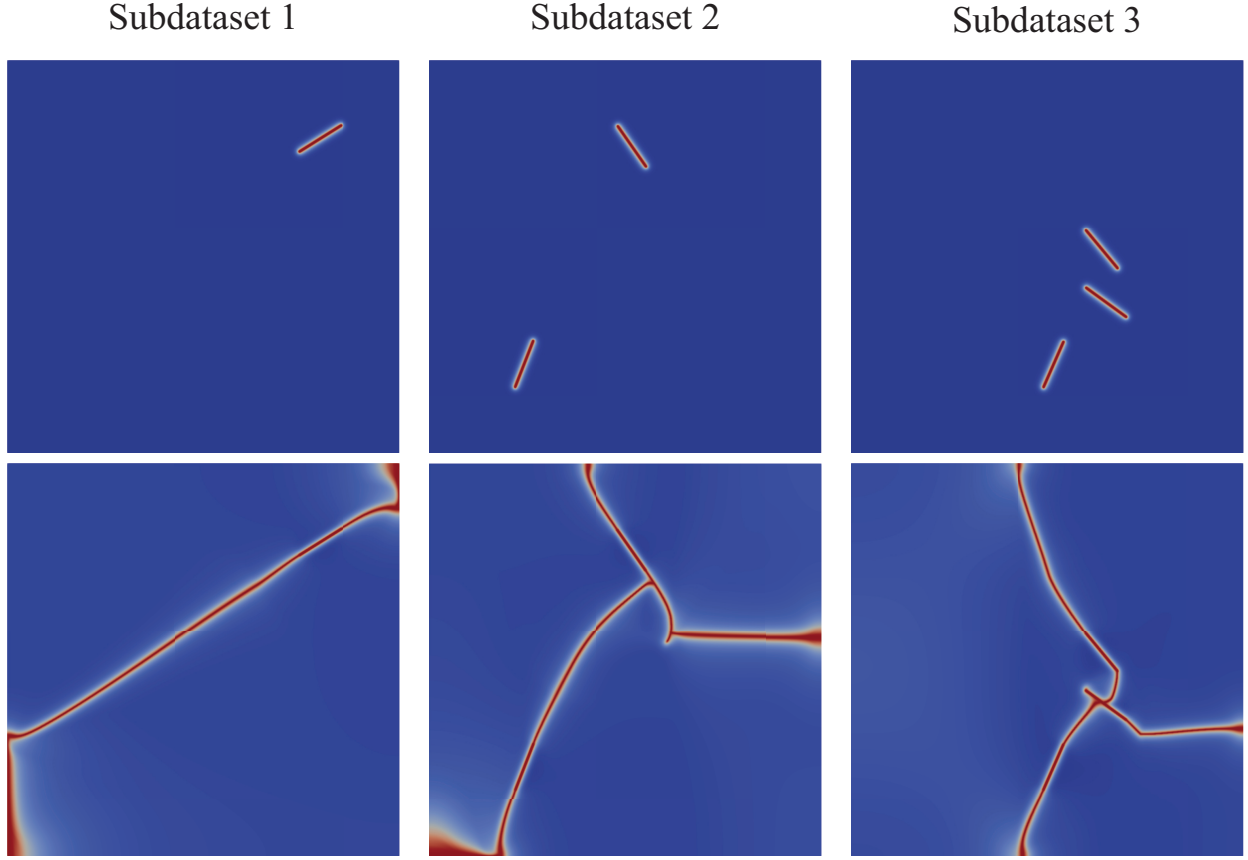


Figure 1: Representative initial crack configurations for the one, two, and three-crack sub-datasets.

1.2 Repository Link

The datasets are accessible via the Harvard Dataverse. You can also use the `download_script` in our Github repository to download the dataset:
github.com/erfanhamdi/pfm_bench